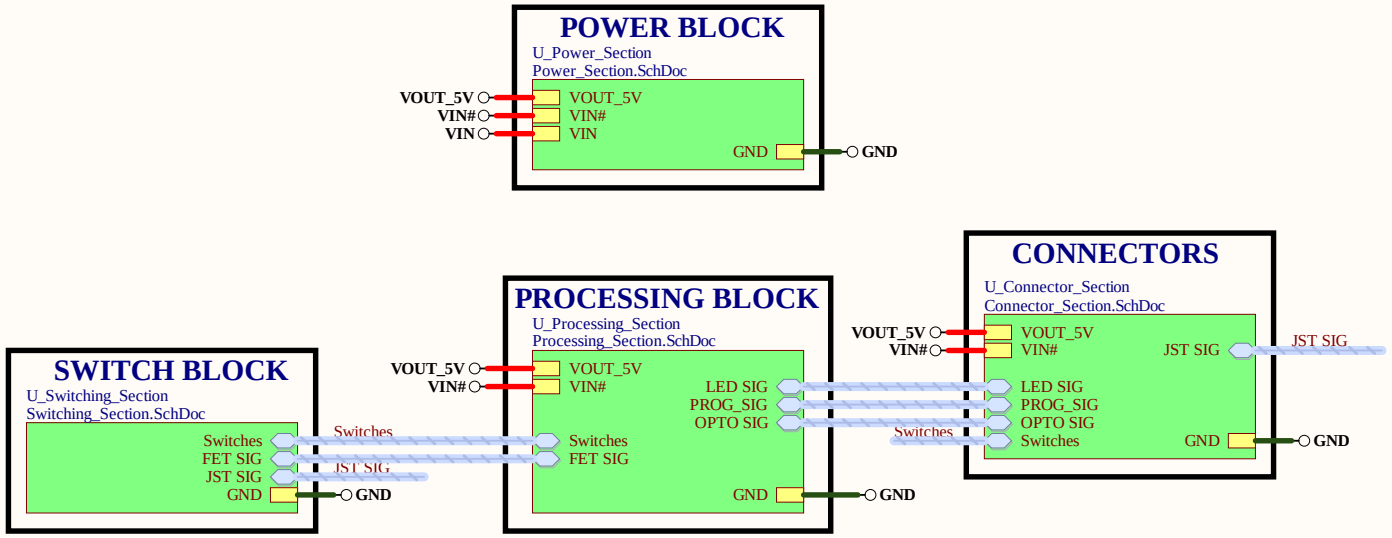
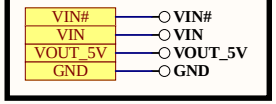
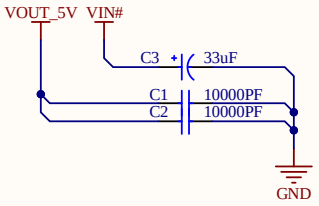

POWER BLOCK
U_Power_Section
Power_Section.SchDoc
VOUT_5V
VIN#
VIN
VOUT_5V
VIN#
VIN
GND
GND
SWITCH BLOCK
U_Switching_Section
Switching_Section.SchDoc
Switches
FET SIG
JST SIG
GND
GND
PROCESSING BLOCK
U_Processing_Section
Processing_Section.SchDoc
VOUT_5V
VIN#
Switches
FET SIG
LED SIG
PROG_SIG
OPTO SIG
GND
GND
CONNECTORS
U_Connector_Section
Connector_Section.SchDoc
VOUT_5V
VIN#
LED SIG
PROG_SIG
OPTO SIG
Switches
JST SIG
GND
GND
JST SIG
Switches

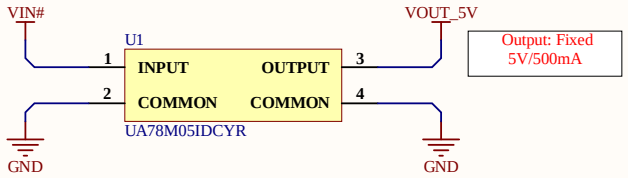
POWER SIGNALS



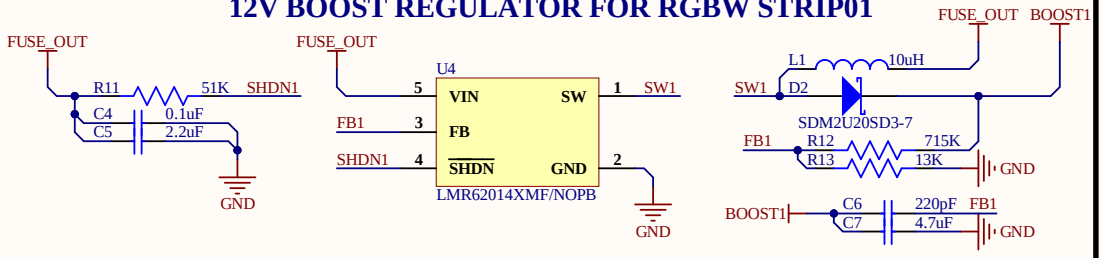
DECOUPLING CAPACITOR



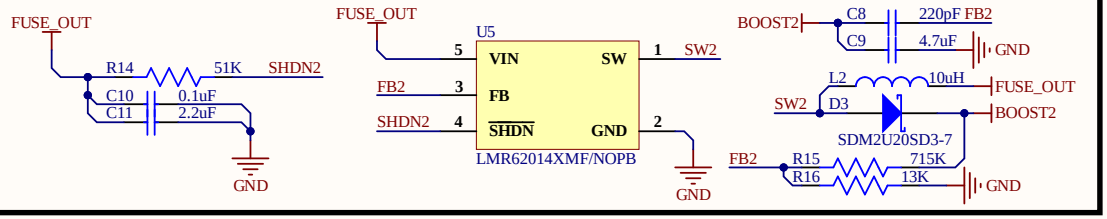
LINEAR REGULATOR



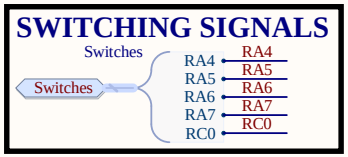
12V BOOST REGULATOR FOR RGBW STRIP01

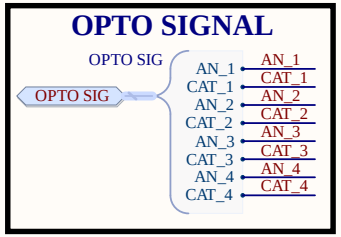


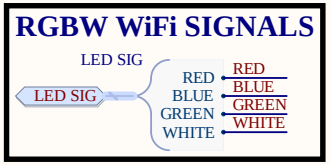
12V BOOST REGULATOR FOR RGBW STRIP02

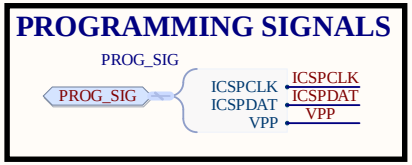


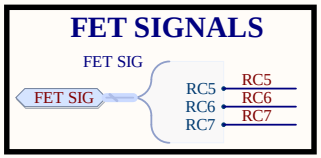
Output: 12V
Switch Current up to 1.4A
R1 = R2 X (VOUT/1.23 − 1)

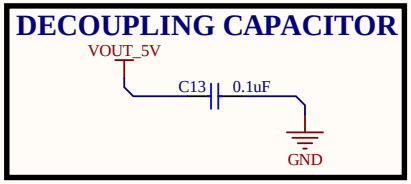

SWITCHING SIGNALS
Switches
Switches
RA4
RA5
RA6
RA7
RC0
RA4
RA5
RA6
RA7
RC0

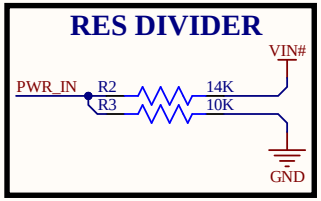

OPTO SIGNAL
OPTO SIG
OPTO SIG
AN_1
CAT_1
AN_2
CAT_2
AN_3
CAT_3
AN_4
CAT_4
AN_1
CAT_1
AN_2
CAT_2
AN_3
CAT_3
AN_4
CAT_4

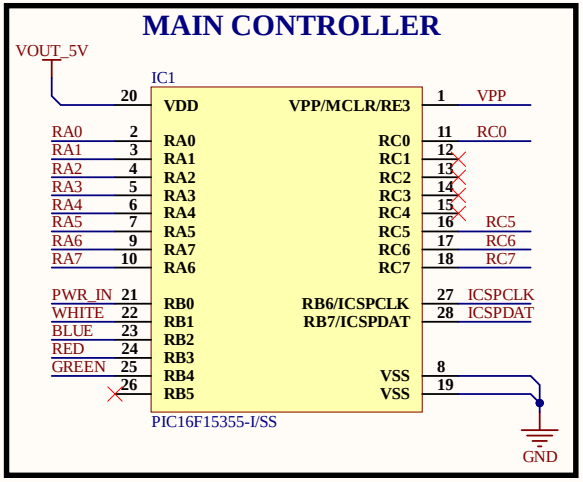

RGBW WiFi SIGNALS
LED SIG
LED SIG
RED
BLUE
GREEN
WHITE
RED
BLUE
GREEN
WHITE

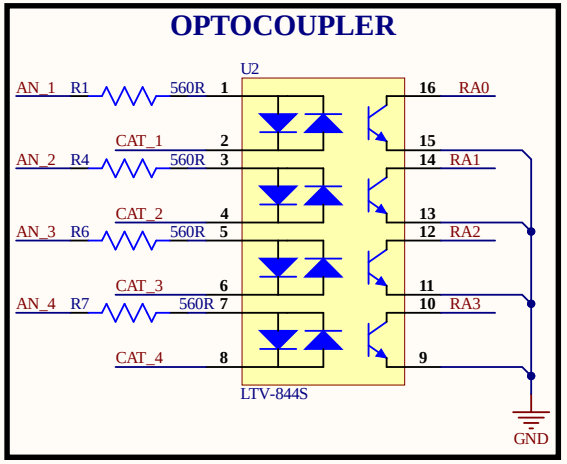

PROGRAMMING SIGNALS
PROG_SIG
PROG_SIG
ICSPCLK
ICSPDAT
VPP
ICSPCLK
ICSPDAT
VPP

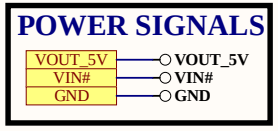

FET SIGNALS
FET SIG
FET SIG
RC5
RC6
RC7
RC5
RC6
RC7

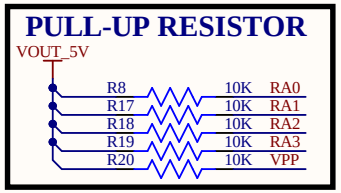

DECOUPLING CAPACITOR
VOUT_5V
C13
0.1uF
GND


RES DIVIDER
VIN#
PWR_IN
R2
14K
R3
10K
GND


MAIN CONTROLLER
VOUT_5V
IC1
20
VDD
VPP/MCLR/RE3
1
VPP
RA0
2
RA0
RC0
11
RC0
RA1
3
RA1
RC1
12
RA2
4
RA2
RC2
13
RA3
5
RA3
RC3
14
RA4
6
RA4
RC4
15
RA5
7
RA5
RC5
16
RC5
RA6
9
RA6
RC6
17
RC6
RA7
10
RA7
RC7
18
RC7
PWR_IN
21
RB0
RB6/ICSPCLK
27
ICSPCLK
WHITE
22
RB1
RB7/ICSPDAT
28
ICSPDAT
BLUE
23
RB2
RED
24
RB3
GREEN
25
RB4
VSS
8
RB5
VSS
19
26
PIC16F15355-I/SS
GND


OPTOCOUPLER
U2
AN_1
R1
560R
1
16
RA0
CAT_1
2
15
AN_2
R4
560R
3
14
RA1
CAT_2
4
13
AN_3
R6
560R
5
12
RA2
CAT_3
6
11
AN_4
R7
560R
7
10
RA3
CAT_4
8
9
LTV-844S
GND

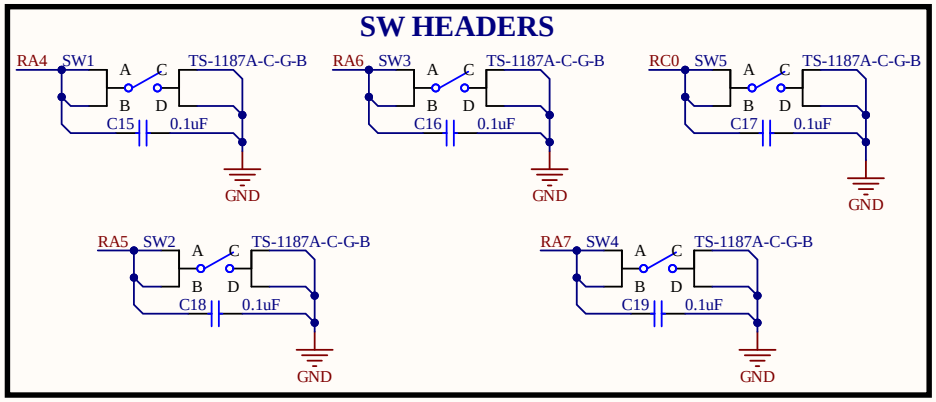

POWER SIGNALS
VOUT_5V
VIN#
GND
VOUT_5V
VIN#
GND


PULL-UP RESISTOR
VOUT_5V
R8
10K
RA0
R17
10K
RA1
R18
10K
RA2
R19
10K
RA3
R20
10K
VPP

SWITCHING SIGNALS



SW HEADERS



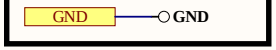
FET SIGNALS



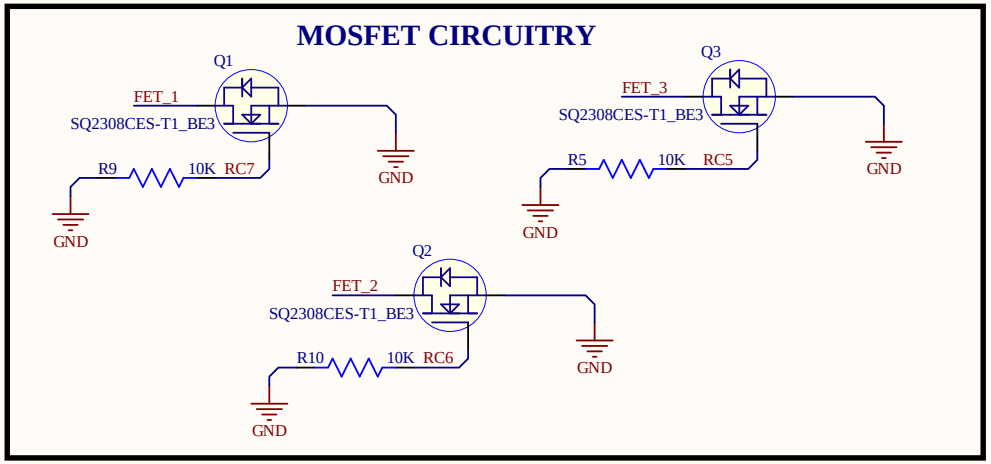
LED OUT SIGNALS



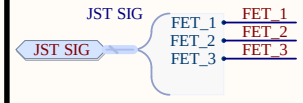
POWER SIGNALS



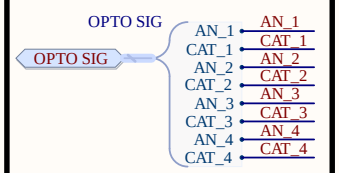
MOSFET CIRCUITRY



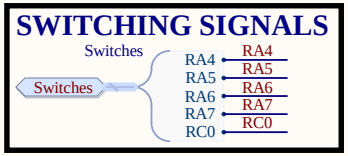
LED OUT SIGNALS



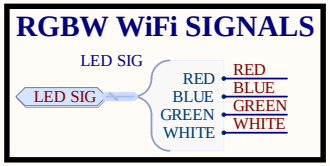
OPTO SIGNAL



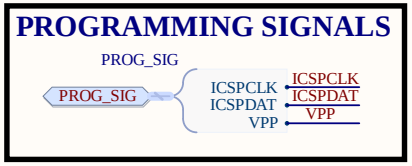
SWITCHING SIGNALS



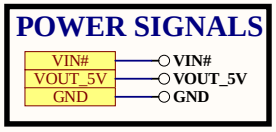
RGBW WiFi SIGNALS



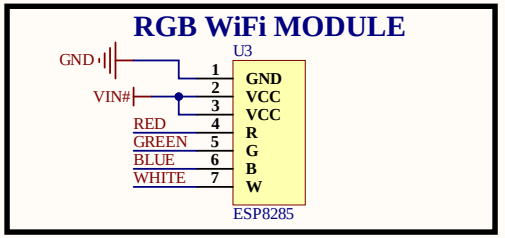
PROGRAMMING SIGNALS



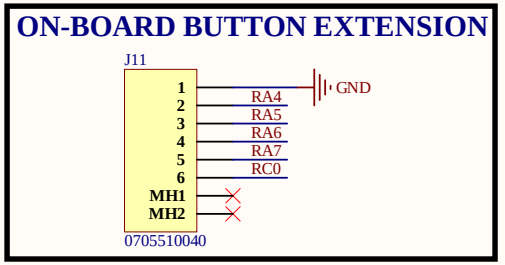
POWER SIGNALS



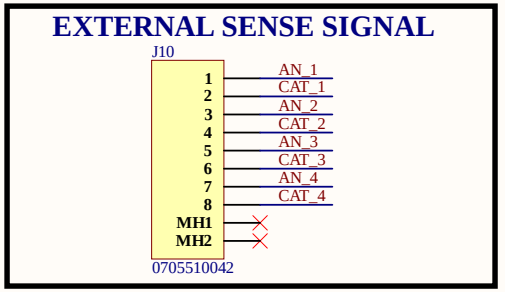
RGB WiFi MODULE



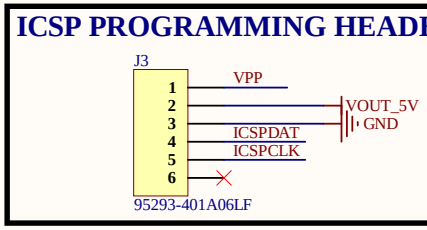
ON-BOARD BUTTON EXTENSION



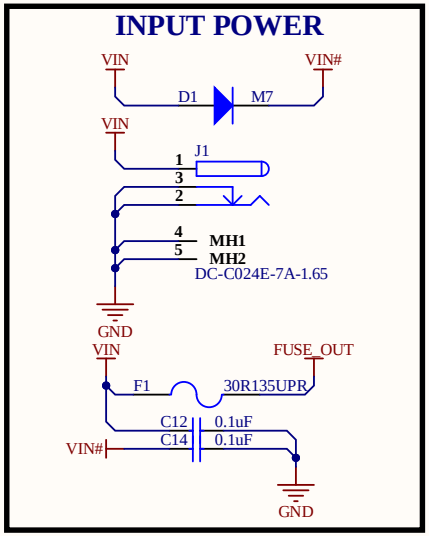
EXTERNAL SENSE SIGNAL



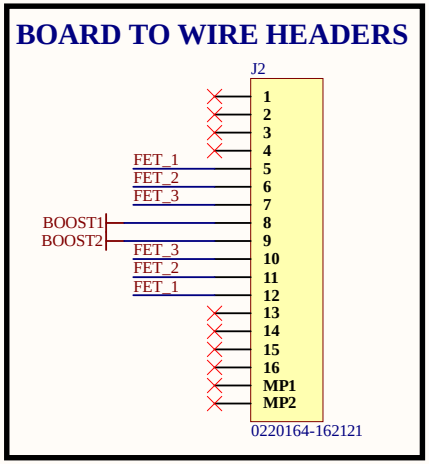
ICSP PROGRAMMING HEADER



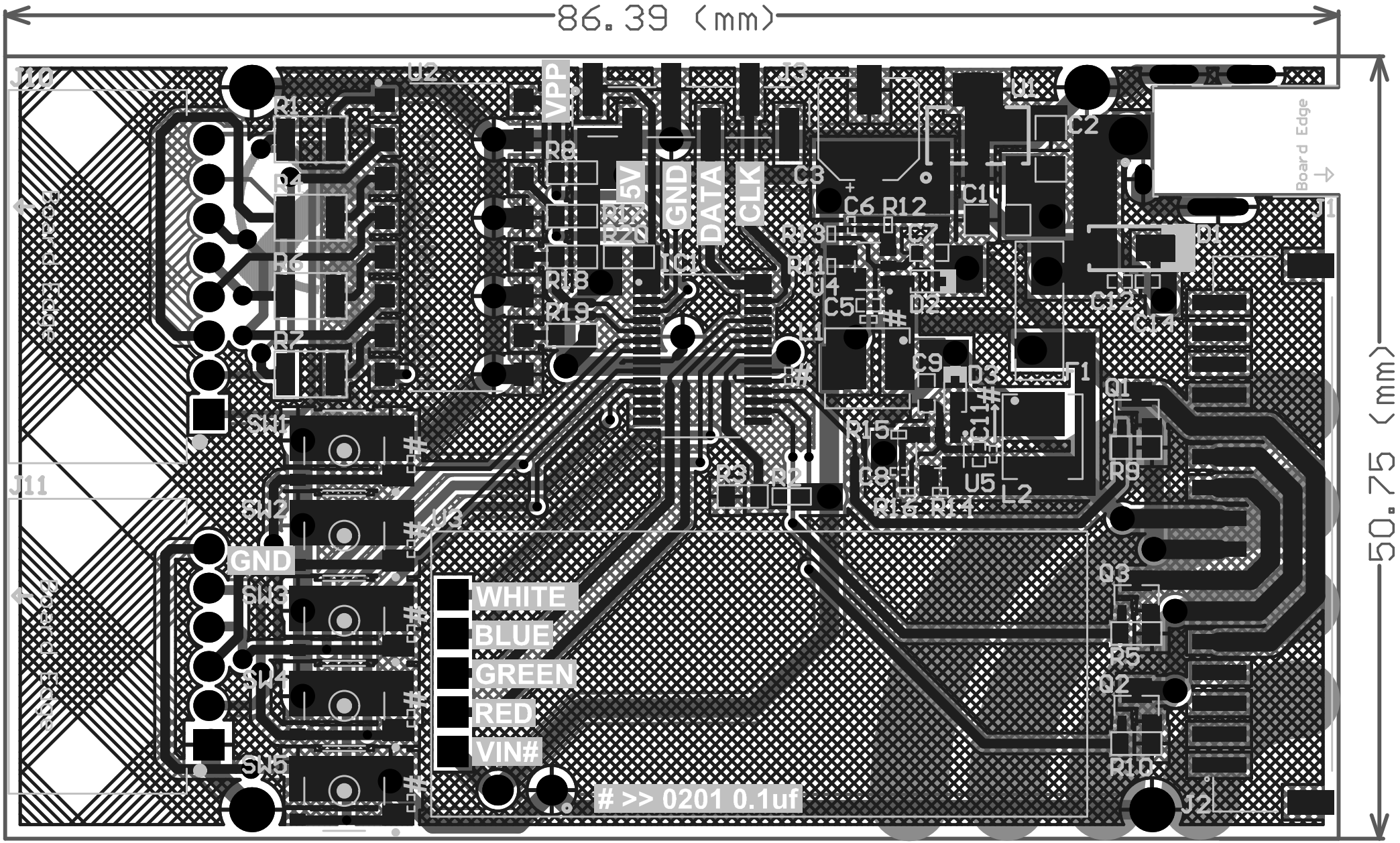
INPUT POWER



BOARD TO WIRE HEADERS



4 pins for each LED strip; VIN, R,G,B signals


86.39 (mm)
50.75 (mm)
VPP
5V
GND
DATA
CLK
WHITE
BLUE
GREEN
RED
VIN#
GND
>> 0201 0.1uf
Board Edge
J10
J11
J12
J13
J1
J2
J3
SW1
SW2
SW3
SW4
SW5
U1
U2
U3
U4
U5
C1
C2
C3
C4
C5
C6
C7
C8
C9
C10
C11
C12
C13
C14
R1
R2
R3
R4
R5
R6
R7
R8
R9
R10
R11
R12
R13
R14
R15
R16
R17
R18
R19
R20
Q1
Q2
Q3
Q4
D1
D2
D3
L1
L2